Chapter P3: Electric circuits

Overview

Known only by its effects, electricity provides an ideal vehicle to illustrate the use and power of scientific models. During the course of the 20th century, electrical engineers completely changed whole societies, by designing systems for electrical generation and distribution, and a whole range of electrical devices.

In this chapter, learners learn how scientists visualise what is going on inside circuits and predict circuit behaviour. Topic P3.1 introduces the idea of electric charge and electric fields. In Topic P3.2, models of charge moving through circuits driven by a voltage and against a resistance are introduced. A more general understanding of voltage as potential difference is then developed in Topic P3.3, which then continues with an exploration of the difference between series and parallel circuits, leading on to investigating the behaviour of various components in d.c. series circuits. Topic P3.4 concentrates on quantifying the energy transferred in electric circuits and how this is determined by both the potential difference and the current.

A reminder of earlier work on magnets and magnetic fields in Topic P3.5 leads into an introduction to the electric motor in Topic P3.6. Applications of electromagnetism and, in particular the electric motor, have revolutionised people's lives in so many ways – from very small motors used in medical contexts, to very large motors used to propel ships or pump water in pumped storage schemes. In Topic P3.7, the process of electromagnetic induction is placed in the context of power generation and the use of transformers in power distribution.

Learning about electric circuits before GCSE (9–1)

From study at Key Stages 1 to 3 learners should:

- be familiar with the basic properties of magnets, and use these to explain and predict observations
- know that there is a magnetic field close to any wire carrying an electric current
- be aware of the existence of electric charge, and understand how simple electrostatic phenomena can be explained in terms of the movement of electrons between and within objects
- understand the idea of an electric circuit (a closed conducting loop containing a battery) that conducts an electric current and be able to predict the current in branches of a parallel circuit
- understand the idea of voltage as a measure of the 'strength' of a battery or power supply
- know that electrical resistance is measured in ohms and can be calculated by dividing the voltage across the component by the current through it
- know that the power ratings of electrical appliances are related to the rate at which the appliances transfers energy.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.

All other statements will be assessed in both Foundation and Higher Tier papers.

P3.1 What is electric charge? (*separate science only*)

Teaching and learning narrative	**Assessable learning outcomes** *Learners will be required to:*
When two objects are rubbed together they become charged, because electrons are transferred from one object to the other. Electrons are negatively charged. Objects with similar charges repel, and objects with opposite charges attract. Around every electric charge there is an electric field; in this region of space the effects of charge can be felt; when another charge enters the field there is an interaction between them and both charges experience a force.	1. describe the production of static electricity, and sparking, by rubbing surfaces, and evidence that charged objects exert forces of attraction or repulsion on one another when not in contact 2. explain how transfer of electrons between objects can explain the phenomenon of static electricity 3. explain the concept of an electric field and how it helps to explain the phenomenon of static electricity

Linked learning opportunities

Practical work

- Demonstrate that there are forces between charged objects and that the effect diminishes with increasing distance between the charges.

P3.2 What determines the current in an electric circuit?

Teaching and learning narrative

An electric current is the rate of flow of charge; in an electric circuit the metal conductors (the components and wires) contain many charges that are free to move. When a circuit is made, the battery causes these free charges to move, and these charges are not used up but flow in a continuous loop.

In a given circuit, the larger the potential difference across the power supply the bigger the current. Components (for example, resistors, lamps, motors) resist the flow of charge through them; the resistance of connecting wires is usually so small that it can be ignored. The larger the resistance in a given circuit, the smaller the current will be.

Representational models of electric circuits use physical analogies to help think about how an electric circuit works, and to predict what happens when a variable is changed (IaS3).

Assessable learning outcomes

Learners will be required to:

1. recall that current is a rate of flow of charge, that for a charge to flow, a source of potential difference and a closed circuit are needed and that a current has the same value at any point in a single closed loop
2. recall and use the relationship between quantity of charge, current and time:
   charge (C) = current (A) × time (s)
   M1c, M3b, M3c, M3d
3. recall that current, I, depends on both the resistance, R, and potential difference, V, and the units in which these quantities are measured
4. a) recall and apply the relationship between I, R, and V, to calculate the currents, potential differences and resistances in d.c. series circuits:
   potential difference (V) = current (A) × resistance (Ω)
   M1c, M3b, M3c, M3d
   b) describe an experiment to investigate the resistance of a wire and be able to draw the circuit diagram of the circuit used
   PAG7
5. recall that for some components the value of R remains constant (fixed resistors) but that in others it can change as the current changes (e.g. heating elements, lamp filaments)
6. a) use graphs to explore whether circuit elements are linear or non-linear and relate the curves produced to their function and properties
   M4c, M4d
   b) describe experiments to investigate the I-V characteristics of circuit elements. To include: lamps, diodes, LDRs and thermistors. Be able to draw circuit diagrams for the circuits used
   PAG6
7. represent circuits with the conventions of positive and negative terminals, and the symbols that represent common circuit elements, filament lamps, diodes, LDRs and thermistors, switches and fixed and variable resistors

Linked learning opportunities

Ideas about Science

- Identify limitations in analogies used to represent electric circuits (IaS3).

Practical work

- Design and construct electric circuits to investigate the electrical properties of range of circuit components.

P3.3 How do series and parallel circuits work?

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*	*Linked learning opportunities*
When electric charge flows through a component (or device), work is done by the power supply and energy is transferred from it to the component and/or its surroundings. Potential difference measures the work done per unit charge. In a series circuit the charge passes through all the components, so the current through each component is the same and the work done on each unit of charge by the battery must equal the total work done by the unit of charge on the components. The potential difference (p.d.) is largest across the component with the greatest resistance and a change in the resistance of one component will result in a change in the potential differences across all the components. In a parallel circuit each charge passes through only one branch of the circuit, so the current through each branch is the same as if it were the only branch present and the work done by each unit of charge is the same for each branch and equal to the work done by the battery on each charge. The current is largest through the component with the smallest resistance, because the same battery p.d. causes a larger current to flow through a smaller resistance than through a bigger one. When two or more resistors are placed in series the effective resistance of the combination (equivalent resistance) is equal to the sum of their resistances, because the battery has to move charges through all of them. Two (or more) resistors in parallel provide more paths for charges to move along than either resistor on its own, so the effective resistance is less. Some components are designed to change resistance in response to changes in the environment e.g. the resistance of an LDR varies with light intensity, the resistance of a thermistor varies with temperature; these properties used in sensing systems to monitor changes in the environment.	1. relate the potential difference between two points in the circuit to the work done on, or by, a given amount of charge as it moves between these points $\text{potential difference (V)} = \dfrac{\text{work done (energy transferred) (J)}}{\text{charge (C)}}$ M1c, M3b, M3c, M3d 2. a) describe the difference between series and parallel circuits: to include ideas about how the current through each component and the potential difference across each component is affected by a change in resistance of a component b) describe how to practically investigate the brightness of bulbs in series and parallel circuits. Be able to draw circuit diagrams for the circuits used *PAG7* 3. explain, why, if two resistors are in series the net resistance is increased, whereas with two in parallel the net resistance is decreased *qualitative only* 4. solve problems for circuits which include resistors in series, using the concept of equivalent resistance M1c, M3b, M3c, M3d 5. explain the design and use of d.c. series circuits for measurement and testing purposes including exploring the effect of: a) changing current in filament lamps, diodes, thermistors and LDRs b) changing light intensity on an LDR c) changing temperature of a thermistor (NTC only)	**Ideas about Scien** • Link the featu of a model or analogy to features in an electric circuit identify evidence for specific aspec of a model an limitations in representatio of a model (IaS3). **Practical work** • Use d.c. series circuits, including potential divid circuits to investigate th behaviour of a variety of components. • Design and construct electric circuit to use a senso for a particula purpose.

P3.4 What determines the rate of energy transfer in a circuit?			
Teaching and learning narrative	**Assessable learning outcomes** *Learners will be required to:*		***Linked learning opportunities***
The energy transferred when electric charge flows through a component (or device), depends on the amount of charge that passes and the potential difference across the component. The power rating (in watts, W) of an electrical device is a measure of the rate at which an electrical power supply transfers energy to the device and/or its surroundings. The rate of energy transfer depends on both the potential difference and the current. The greater the potential difference, the faster the charges move through the circuit, and the more energy each charge transfers. The National Grid uses transformers to step down the current for power transmission. The power output from a transformer cannot be greater than the power input, therefore if the current increases, the potential difference must decrease. Transmitting power with a lower current through the cables results in less power being dissipated during transmission.	1.	describe the energy transfers that take place when a system is changed by work done when a current flows through a component	**Practical work** • Compare the power consumption of a variety of devices and relate it to the current passing through the device.
	2.	explain, with reference to examples, how the power transfer in any circuit device is related to the energy transferred from the power supply to the device and its surroundings over a given time: $\text{power (W)} = \dfrac{\text{energy transferred (J)}}{\text{time (s)}}$ M1c, M3b, M3c, M3d	
	3.	recall and use the relationship between the potential difference across the component and the total charge to calculate the energy transferred in an electric circuit when a current flows through a component: energy transferred (work done) (J) = charge (C) × potential difference (V) M1c, M3b, M3c, M3d	
	4.	recall and apply the relationships between power transferred in any circuit device, the potential difference across it, the current through it, and its resistance: a) power (W) = potential difference (V) × current (A) b) power (W) = (current (A))^2 × resistance (Ω) M1c, M3b, M3c, M3d	
	5.	use the idea of conservation of energy to show that when a transformer steps up the voltage, the output current must decrease and vice versa. Select and use the equation: potential difference across primary coil (V) × current in primary coil (A) = potential difference across secondary coil (V) × current in secondary coil (A) M1c, M3b, M3c, M3d	
	6.	explain how transmitting power at higher voltages is more efficient way to transfer energy	

P3.5 What are magnetic fields?

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*
Around any magnet there is a region, called the magnetic field, in which another magnet experiences a force. The magnetic effect is strongest at the poles. The field gets gradually weaker with distance from the magnet. The direction and strength of a magnetic field can be represented by field lines. These show the direction of the force that would be experienced by the N pole of a small magnet, placed in the field. The magnetic field around the Earth, with poles near the geographic north and south, provides evidence that the core of the Earth is magnetic. The N-pole of a magnetic compass will point towards the magnetic north pole. Magnetic materials (such as iron and nickel) can be induced to become magnets by placing them in a magnetic field. When the field is removed permanent magnets retain their magnetisation whilst other materials lose their magnetisation. When there is an electric current in a wire, there is a magnetic field around the wire; the field lines form concentric circles around the wire. Winding the wire into a coil (solenoid) makes the magnetic field stronger, as the fields of each turn add together. Winding the coil around an iron core makes a stronger magnetic field and an electromagnet that can be switched on and off. In loudspeakers and headphones the magnetic field produced due to a current through a coil interacts with the field of a permanent magnet. The 19th century discovery of this electromagnetic effect led quickly to the invention of a number of magnetic devices, including electromagnetic relays, which formed the basis of the telegraph system, leading to a communications revolution (IaS4.1).	1. describe the attraction and repulsion between unlike and like poles for permanent magnets 2. describe the characteristics of the magnetic field of a magnet, showing how strength and direction change from one point to another 3. explain how the behaviour of a magnetic compass is related to evidence that the core of the Earth must be magnetic 4. describe the difference between permanent and induced magnets 5. describe how to show that a current can create a magnetic effect 6. describe the pattern and directions of the magnetic field around a conducting wire 7. recall that the strength of the field depends on the current and the distance from the conductor 8. explain how the magnetic effect of a solenoid can be increased **9. explain how a solenoid can be used to generate sound in loudspeakers and headphones** *(separate science only)*

Linked learning opportunities

Specification links

- Sound waves (P1.4).

Practical work

- Use plotting compasses to map the magnetic field near a permanent bar magnet, between facing like/opposite poles of two magnets, a single wire, a flat coil of wire and a solenoid.
- Investigate the relationship between the number of turns on a solenoid and the strength of the magnetic field.
- **Build a loudspeaker.**

Ideas about Science

- Developments of electromagnets have led to major changes in people's lives, including applications in communications systems, MRI scanners and on cranes in scrapyards.

P3.6 How do electric motors work?

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*
The magnetic fields of a current-carrying wire and a nearby permanent magnet will interact and the wire and magnet exert a force on each other. This is called the 'motor effect'.	**1. describe the interaction forces between a magnet and a current-carrying conductor to include ideas about magnetic fields**
If the current-carrying wire is placed at right angles to the magnetic field lines, the force will be at right angles to both the current direction and the lines of force of the field. The direction of the force can be inferred using Fleming's left-hand rule.	**2. show that Fleming's left-hand rule represents the relative orientations of the force, the conductor and the magnetic field**
The size of the force is proportional to the length of wire in the field, the current and the strength of the field.	**3. select and apply the equation that links the force, *F*, on a conductor to the strength of the field, *B*, the size of the current, *I*, and the length of conductor, *l*, to calculate the forces involved** **force (N) = magnetic flux density (T) × current (A) × length of conductor (m)** M1b, M1c, M3b, M3c, M3d
The motor effect can result in a turning force on a rectangular current-carrying coil placed in a uniform magnetic field; this is the principle behind all electric motors. **The invention and development of practical electric motors have made an impact on almost every aspect of daily life (IaS4.1).**	**4. explain how the force on a conductor in a magnetic field is used to cause rotation in the rectangular coil of a simple electric motor** ⓘ ***Detailed knowledge of the construction of motors not required***

Linked learning opportunities

Practical work

- Investigate the motor effect for a single wire in a magnetic field and apply the principles to build a simple electric motor.
- Build a simple electric motor and explain how it works.

Ideas about Science

- Describe and explain examples of uses of electric motors that have made significant improvements to people's lives. (IaS4.1).

P3.7 What is the process inside an electric generator? (*separate science only*)

Teaching and learning narrative

Mains electricity is produced using the process of electromagnetic induction.

When a magnet is moving into a coil of wire a potential difference is induced across the ends of the coil; if the magnet is moving out of the coil, or the other pole of the magnet is moving into it, there is a potential difference induced in the opposite direction. If the ends of the coil are connected to make a closed circuit, a current will flow round the circuit.

In a moving coil microphone sound waves cause a diaphragm to vibrate. The diaphragm is attached to a coil which is in the field of a permanent magnet. Sounds make the coil vibrate, inducing a changing potential difference across the ends of the coil. This potential difference drives a changing current in an electric circuit.

In a generator, a magnet or electromagnet is rotated within a coil of wire to induce a voltage across the ends of the coil.

The induced voltage across the coil of an alternating current (a.c.) generator (and hence the current in an external circuit) changes during each revolution of the magnet or electromagnet.

To generate a d.c. split-ring commutator is used so that the current always passes from the same side of the generator.

Assessable learning outcomes

Learners will be required to:

1. **recall that a change in the magnetic field around a conductor can give rise to an induced potential difference across its ends, which could drive a current**
2. **explain the action of a moving coil microphone in converting the pressure variations in sound waves into variations in current in electrical circuits**
3. **recall that the direction of the induced potential difference drives a current which generates a second magnetic field that would oppose the original change in field**
4. **use ideas about electromagnetic induction to explain a potential difference/time graph showing the output from an alternator being used to generate a.c.**
5. **explain how an alternator can be adapted to produce a dynamo to generate d.c., including explaining a potential difference/time graph**
6. **explain how the effect of an alternating current in one circuit in inducing a current in another is used in transformers**
7. **describe how the ratio of the potential differences across the two coils in a transformer depends on the ratio of the numbers of turns in each**

Linked learning opportunities

Specification links

- How can electricity be generated? (P2.2
- Sound waves (P1.4)

Practical work

- Investigate electromagnetic induction in transformers and generators.

Ideas about Science

- Describe and explain examples o technological applications of science that have made significant positive differences to people's lives (IaS4).
- Identify examples of risks which arise from a new scientific or technological advance (IaS4).

P3.7 What is the process inside an electric generator? (*separate science only*)

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*	*Linked learning opportunities*
A changing magnetic field caused by changes in the current in one coil of wire can induce a voltage in a neighbouring coil. **A simple transformer has two coils of wire wound on an iron core; a changing current in one coil of a transformer will cause a changing magnetic field in the iron core, which in turn will induce a changing potential difference across the other transformer coil.** **The discovery of electromagnetic induction and the subsequent development of power generators transformed the way we live, although with new developments in technology there are often unintended consequences (IaS4).**	**8. apply the equations linking the potential differences and numbers of turns in the two coils of a transformer, to the currents and the power transfer involved and relate these to the advantages of power transmission at high voltages.** **a) potential difference across primary coil (V) × current in primary coil (A) = potential difference across secondary coil (V) × current in secondary coil (A)** **b)** $\dfrac{\text{potential difference across primary coil (V)}}{\text{potential difference across secondary coil (V)}} = \dfrac{\text{number of turns in primary coil}}{\text{number of turns in secondary coil}}$ M1c, M3b, M3c	